

STA314 Assignment 2

November 12, 2023

```
[1]: library(tidyverse)
```

```
Attaching core tidyverse packages          tidyverse
2.0.0
dplyr      1.1.2      readr      2.1.4
forcats    1.0.0      stringr   1.5.0
ggplot2    3.4.2      tibble    3.2.1
lubridate  1.9.2      tidyr     1.3.0
purrr      1.0.1
Conflicts
tidyverse_conflicts()
dplyr::filter() masks stats::filter()
dplyr::lag()     masks stats::lag()
Use the conflicted package
(<http://conflicted.r-lib.org/>) to force all conflicts to
become errors
```

```
[2]: library(ggplot2)
```

```
[3]: library(readxl)
```

```
[4]: library(e1071)
install.packages("caret")
library(caret)
install.packages("glmnet")
library(glmnet)
```

```
Installing package into '/opt/r'
(as 'lib' is unspecified)
```

```
also installing the dependencies 'shape', 'progressr', 'SQUAREM', 'diagram',
'lava', 'rpart', 'nnet', 'prodlim', 'iterators', 'clock', 'gower', 'hardhat',
'ipred', 'timeDate', 'foreach', 'ModelMetrics', 'pROC', 'recipes'
```

```
Loading required package: ggplot2
```

```
Loading required package: lattice
```

Installing package into '/opt/r'
(as 'lib' is unspecified)

Loading required package: Matrix

Loaded glmnet 4.1-8

```
[5]: df = read_excel("EMV_VIX_Data.xlsx")
```

```
[6]: data <- read_excel("EMV_VIX_Data.xlsx")
y = data$VIX
data <- data[,!(names(data)%in%c("VIX", "Date"))]

x = model.matrix(y~., data)[,-1]
```

OLS Regression

```
[7]: ols = lm(y~x)
summary(ols)
```

Call:

```
lm(formula = y ~ x)
```

Residuals:

Min	1Q	Median	3Q	Max
-12.7853	-3.0176	-0.5303	2.3708	19.7427

Coefficients:

	Estimate
(Intercept)	9.013772
xEMV	0.250543
x`Political Uncertainty Tracker`	0.544205
x`Infectious Disease EMV Tracker`	0.155782
x`Macroeconomic News and Outlook EMV Tracker`	0.330970
x`Macro - Broad Quantity Indicators EMV Tracker`	0.529192
x`Macro - Inflation EMV Indicator`	1.038123
x`Macro - Interest Rates EMV Tracker`	-1.432616
x`Macro - Other Financial Indicators EMV Tracker`	-0.444125
x`Macro - Labor Markets EMV Tracker`	0.130995
x`Macro - Real Estate Markets EMV Tracker`	-0.121219
x`Macro - Trade EMV Tracker`	-2.277171
x`Macro - Business Investment and Sentiment EMV Tracker`	0.945355
x`Macro - Consumer Spending and Sentiment EMV Tracker`	-0.019194
x`Commodity Markets EMV Tracker`	-0.128484
x`Financial Crises EMV Tracker`	0.104002

x`Exchange Rates EMV Tracker`	0.351962
x`Healthcare Matters EMV Tracker`	-1.041695
x`Litigation Matters EMV Tracker`	-1.313388
x`Competition Matters EMV Tracker`	0.107008
x`Labor Disputes EMV Tracker`	3.126042
x`Intellectual Property Matters EMV Tracker`	2.937708
x`Fiscal Policy EMV Tracker`	0.477432
x`Taxes EMV Tracker`	-0.774153
x`Government Spending, Deficits, and Debt EMV Tracker`	-1.187397
x`Entitlement and Welfare Programs EMV Tracker`	0.438513
x`Monetary Policy EMV Tracker`	-0.490020
x`Regulation EMV Tracker`	-1.045138
x`Financial Regulation EMV Tracker`	1.703597
x`Competition Policy EMV Tracker`	0.850412
x`Intellectual Property Policy EMV Tracker`	-0.907107
x`Labor Regulations EMV Tracker`	0.353459
x`Immigration EMV Tracker`	-1.052707
x`Energy and Environmental Regulation EMV Tracker`	3.736002
x`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	1.394200
x`Housing and Land Management EMV Tracker`	0.014570
x`Other Regulation EMV Tracker`	-3.875063
x`National Security Policy EMV Tracker`	-0.078670
x`Government-Sponsored Enterprises EMV Tracker`	0.540309
x`Trade Policy EMV Tracker`	-0.540150
x`Healthcare Policy EMV Tracker`	-0.039388
x`Food and Drug Policy EMV Tracker`	0.994212
x`Transportation, Infrastructure, and Public Utilities EMV Tracker`	-0.481211
x`Elections and Political Governance EMV Tracker`	-0.004546
x`Agricultural Policy EMV Tracker`	-7.021830
x`Petroleum Markets EMV Tracker`	-0.084434
	Std. Error
(Intercept)	1.141343
xEMV	0.392186
x`Political Uncertainty Tracker`	0.511832
x`Infectious Disease EMV Tracker`	0.118871
x`Macroeconomic News and Outlook EMV Tracker`	0.510646
x`Macro - Broad Quantity Indicators EMV Tracker`	0.260862
x`Macro - Inflation EMV Indicator`	0.259375
x`Macro - Interest Rates EMV Tracker`	0.298730
x`Macro - Other Financial Indicators EMV Tracker`	1.053383
x`Macro - Labor Markets EMV Tracker`	0.326937
x`Macro - Real Estate Markets EMV Tracker`	0.276512
x`Macro - Trade EMV Tracker`	1.433143
x`Macro - Business Investment and Sentiment EMV Tracker`	1.172565
x`Macro - Consumer Spending and Sentiment EMV Tracker`	0.494838
x`Commodity Markets EMV Tracker`	0.475116
x`Financial Crises EMV Tracker`	0.251963
x`Exchange Rates EMV Tracker`	0.656692

x`Healthcare Matters EMV Tracker`	1.105969
x`Litigation Matters EMV Tracker`	1.068237
x`Competition Matters EMV Tracker`	0.893026
x`Labor Disputes EMV Tracker`	0.773382
x`Intellectual Property Matters EMV Tracker`	1.106260
x`Fiscal Policy EMV Tracker`	0.638014
x`Taxes EMV Tracker`	0.551763
x`Government Spending, Deficits, and Debt EMV Tracker`	0.437387
x`Entitlement and Welfare Programs EMV Tracker`	0.598358
x`Monetary Policy EMV Tracker`	0.272027
x`Regulation EMV Tracker`	0.486112
x`Financial Regulation EMV Tracker`	0.604194
x`Competition Policy EMV Tracker`	0.860941
x`Intellectual Property Policy EMV Tracker`	6.517487
x`Labor Regulations EMV Tracker`	1.241833
x`Immigration EMV Tracker`	2.616309
x`Energy and Environmental Regulation EMV Tracker`	1.482296
x`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	1.533402
x`Housing and Land Management EMV Tracker`	1.422814
x`Other Regulation EMV Tracker`	1.800489
x`National Security Policy EMV Tracker`	0.195414
x`Government-Sponsored Enterprises EMV Tracker`	0.442987
x`Trade Policy EMV Tracker`	0.355949
x`Healthcare Policy EMV Tracker`	1.426481
x`Food and Drug Policy EMV Tracker`	1.849548
x`Transportation, Infrastructure, and Public Utilities EMV Tracker`	1.774003
x`Elections and Political Governance EMV Tracker`	0.347639
x`Agricultural Policy EMV Tracker`	3.981032
x`Petroleum Markets EMV Tracker`	0.343567
	t value
(Intercept)	7.898
xEMV	0.639
x`Political Uncertainty Tracker`	1.063
x`Infectious Disease EMV Tracker`	1.311
x`Macroeconomic News and Outlook EMV Tracker`	0.648
x`Macro - Broad Quantity Indicators EMV Tracker`	2.029
x`Macro - Inflation EMV Indicator`	4.002
x`Macro - Interest Rates EMV Tracker`	-4.796
x`Macro - Other Financial Indicators EMV Tracker`	-0.422
x`Macro - Labor Markets EMV Tracker`	0.401
x`Macro - Real Estate Markets EMV Tracker`	-0.438
x`Macro - Trade EMV Tracker`	-1.589
x`Macro - Business Investment and Sentiment EMV Tracker`	0.806
x`Macro - Consumer Spending and Sentiment EMV Tracker`	-0.039
x`Commodity Markets EMV Tracker`	-0.270
x`Financial Crises EMV Tracker`	0.413
x`Exchange Rates EMV Tracker`	0.536
x`Healthcare Matters EMV Tracker`	-0.942

x`Litigation Matters EMV Tracker`	-1.229
x`Competition Matters EMV Tracker`	0.120
x`Labor Disputes EMV Tracker`	4.042
x`Intellectual Property Matters EMV Tracker`	2.656
x`Fiscal Policy EMV Tracker`	0.748
x`Taxes EMV Tracker`	-1.403
x`Government Spending, Deficits, and Debt EMV Tracker`	-2.715
x`Entitlement and Welfare Programs EMV Tracker`	0.733
x`Monetary Policy EMV Tracker`	-1.801
x`Regulation EMV Tracker`	-2.150
x`Financial Regulation EMV Tracker`	2.820
x`Competition Policy EMV Tracker`	0.988
x`Intellectual Property Policy EMV Tracker`	-0.139
x`Labor Regulations EMV Tracker`	0.285
x`Immigration EMV Tracker`	-0.402
x`Energy and Environmental Regulation EMV Tracker`	2.520
x`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	0.909
x`Housing and Land Management EMV Tracker`	0.010
x`Other Regulation EMV Tracker`	-2.152
x`National Security Policy EMV Tracker`	-0.403
x`Government-Sponsored Enterprises EMV Tracker`	1.220
x`Trade Policy EMV Tracker`	-1.517
x`Healthcare Policy EMV Tracker`	-0.028
x`Food and Drug Policy EMV Tracker`	0.538
x`Transportation, Infrastructure, and Public Utilities EMV Tracker`	-0.271
x`Elections and Political Governance EMV Tracker`	-0.013
x`Agricultural Policy EMV Tracker`	-1.764
x`Petroleum Markets EMV Tracker`	-0.246
	Pr(> t)
(Intercept)	3.74e-14
xEMV	0.52335
x`Political Uncertainty Tracker`	0.28840
x`Infectious Disease EMV Tracker`	0.19088
x`Macroeconomic News and Outlook EMV Tracker`	0.51732
x`Macro - Broad Quantity Indicators EMV Tracker`	0.04325
x`Macro - Inflation EMV Indicator`	7.66e-05
x`Macro - Interest Rates EMV Tracker`	2.41e-06
x`Macro - Other Financial Indicators EMV Tracker`	0.67356
x`Macro - Labor Markets EMV Tracker`	0.68891
x`Macro - Real Estate Markets EMV Tracker`	0.66138
x`Macro - Trade EMV Tracker`	0.11298
x`Macro - Business Investment and Sentiment EMV Tracker`	0.42066
x`Macro - Consumer Spending and Sentiment EMV Tracker`	0.96908
x`Commodity Markets EMV Tracker`	0.78699
x`Financial Crises EMV Tracker`	0.68003
x`Exchange Rates EMV Tracker`	0.59233
x`Healthcare Matters EMV Tracker`	0.34690
x`Litigation Matters EMV Tracker`	0.21972

x`Competition Matters EMV Tracker`	0.90469
x`Labor Disputes EMV Tracker`	6.52e-05
x`Intellectual Property Matters EMV Tracker`	0.00828
x`Fiscal Policy EMV Tracker`	0.45478
x`Taxes EMV Tracker`	0.16149
x`Government Spending, Deficits, and Debt EMV Tracker`	0.00696
x`Entitlement and Welfare Programs EMV Tracker`	0.46413
x`Monetary Policy EMV Tracker`	0.07251
x`Regulation EMV Tracker`	0.03224
x`Financial Regulation EMV Tracker`	0.00508
x`Competition Policy EMV Tracker`	0.32395
x`Intellectual Property Policy EMV Tracker`	0.88939
x`Labor Regulations EMV Tracker`	0.77610
x`Immigration EMV Tracker`	0.68766
x`Energy and Environmental Regulation EMV Tracker`	0.01217
x`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	0.36386
x`Housing and Land Management EMV Tracker`	0.99184
x`Other Regulation EMV Tracker`	0.03206
x`National Security Policy EMV Tracker`	0.68750
x`Government-Sponsored Enterprises EMV Tracker`	0.22340
x`Trade Policy EMV Tracker`	0.13005
x`Healthcare Policy EMV Tracker`	0.97799
x`Food and Drug Policy EMV Tracker`	0.59124
x`Transportation, Infrastructure, and Public Utilities EMV Tracker`	0.78635
x`Elections and Political Governance EMV Tracker`	0.98957
x`Agricultural Policy EMV Tracker`	0.07864
x`Petroleum Markets EMV Tracker`	0.80601

(Intercept)

xEMV

x`Political Uncertainty Tracker`	
x`Infectious Disease EMV Tracker`	
x`Macroeconomic News and Outlook EMV Tracker`	
x`Macro - Broad Quantity Indicators EMV Tracker`	*
x`Macro - Inflation EMV Indicator`	***
x`Macro - Interest Rates EMV Tracker`	***
x`Macro - Other Financial Indicators EMV Tracker`	
x`Macro - Labor Markets EMV Tracker`	
x`Macro - Real Estate Markets EMV Tracker`	
x`Macro - Trade EMV Tracker`	
x`Macro - Business Investment and Sentiment EMV Tracker`	
x`Macro - Consumer Spending and Sentiment EMV Tracker`	
x`Commodity Markets EMV Tracker`	
x`Financial Crises EMV Tracker`	
x`Exchange Rates EMV Tracker`	
x`Healthcare Matters EMV Tracker`	
x`Litigation Matters EMV Tracker`	
x`Competition Matters EMV Tracker`	

```

x`Labor Disputes EMV Tracker`          ***
x`Intellectual Property Matters EMV Tracker`  **
x`Fiscal Policy EMV Tracker`
x`Taxes EMV Tracker`
x`Government Spending, Deficits, and Debt EMV Tracker`  **
x`Entitlement and Welfare Programs EMV Tracker`
x`Monetary Policy EMV Tracker`          .
x`Regulation EMV Tracker`              *
x`Financial Regulation EMV Tracker`      **
x`Competition Policy EMV Tracker`
x`Intellectual Property Policy EMV Tracker`
x`Labor Regulations EMV Tracker`
x`Immigration EMV Tracker`
x`Energy and Environmental Regulation EMV Tracker`      *
x`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`
x`Housing and Land Management EMV Tracker`
x`Other Regulation EMV Tracker`          *
x`National Security Policy EMV Tracker`
x`Government-Sponsored Enterprises EMV Tracker`
x`Trade Policy EMV Tracker`
x`Healthcare Policy EMV Tracker`
x`Food and Drug Policy EMV Tracker`
x`Transportation, Infrastructure, and Public Utilities EMV Tracker`
x`Elections and Political Governance EMV Tracker`
x`Agricultural Policy EMV Tracker`      .
x`Petroleum Markets EMV Tracker`

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.991 on 349 degrees of freedom

Multiple R-squared: 0.6214, Adjusted R-squared: 0.5725

F-statistic: 12.73 on 45 and 349 DF, p-value: < 2.2e-16

```

[43]: predicted_values <- predict(ols)
MSE = mean((y - predicted_values)^2)
MSE
RMSE(predicted_values, y)
vif = 1/(1-0.6214)
vif

```

22.0113727453303

4.6916279419121

2.64131008980454

```

[10]: install.packages("ISLR")
library(ISLR)

```

```
data <- read_excel("EMV_VIX_Data.xlsx")
y = data$VIX
data <- data[,!(names(data)%in%c("VIX", "Date"))]

x = model.matrix(y~., data)[,-1]
```

Installing package into ‘/opt/r’
(as ‘lib’ is unspecified)

Ridge Regression

```
[11]: set.seed(1)
lasso.cv = cv.glmnet(x, y, type.measure="mse", alpha=0, family="gaussian",
  ↪ nlambda=200, nfolds = 10)
```

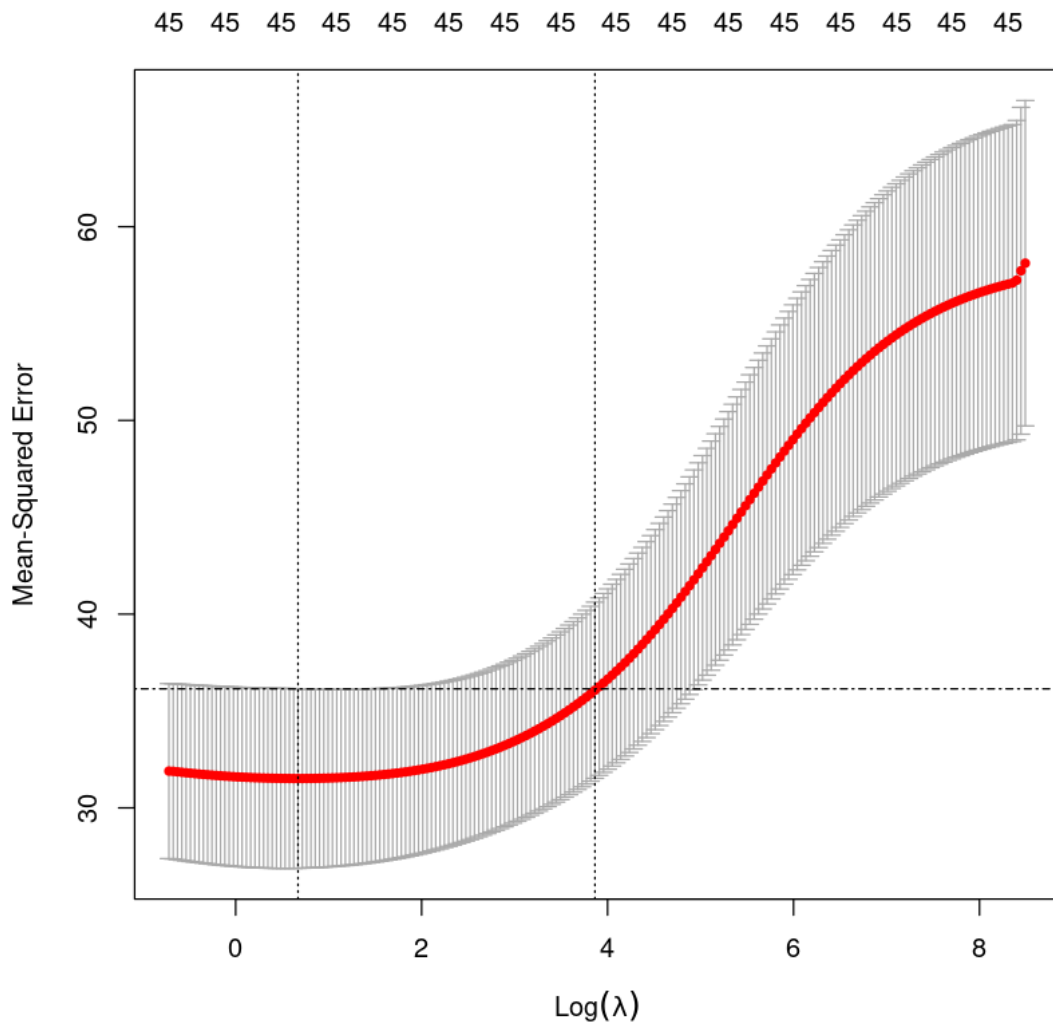
```
[12]: lasso.cv$lambda.min
```

1.95583427263634

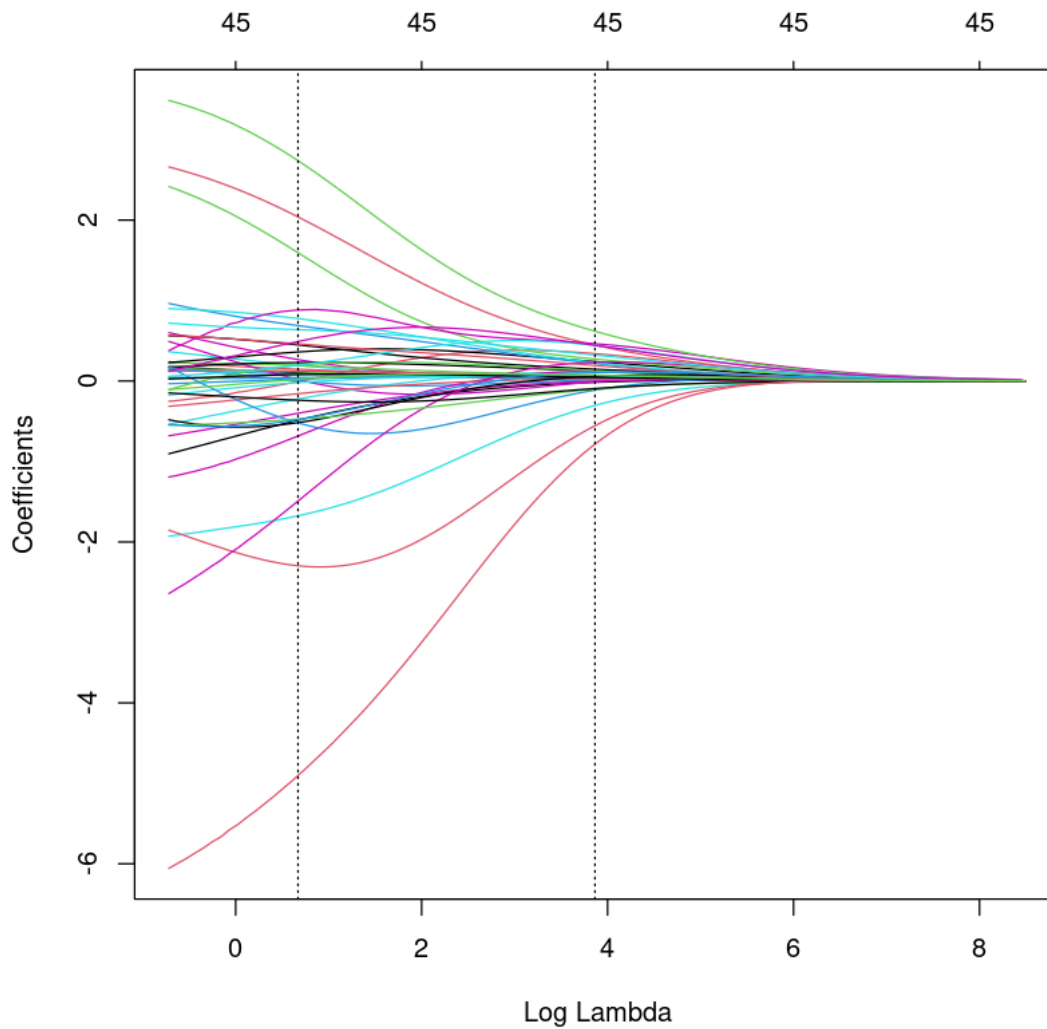
```
[13]: lasso.cv$lambda.1se
```

47.6723680644601

```
[14]: plot(lasso.cv)
abline( h = lasso.cv$cvup[lasso.cv$index[1]], lty = 4 )
```

```
[15]: lasso = glmnet(x, y, alpha = 0)
      plot(lasso, xvar = 'lambda')
      abline(v = log(lasso.cv$lambda.min), lty = 3) # careful to use the log here
      abline(v = log(lasso.cv$lambda.1se), lty = 3)
```



Finding MSE, R Squared, RMSE, VIF

```
[16]: lasso = glmnet(x, y, alpha = 0)
predicted_values = predict(lasso, newx = x, s = lasso.cv$lambda.min)
MSE = mean((y - predicted_values)^2)
MSE
SSE = sum((y - predicted_values)^2)
SST = sum((y - mean(y))^2)
R_squared = 1 - (SSE/SST)
R_squared
RMSE = RMSE(predicted_values, y)
RMSE
VIF = 1/(1-R_squared)
```

VIF

24.7088313344238

0.574964757936285

4.97079785692638

2.35274608087697

Check Coefficients

[17]: `coef(lasso, lasso.cv$lambda.min)`

46 x 1 sparse Matrix of class "dgCMatrix"

	s1
(Intercept)	8.378027460
EMV	0.104847874
`Political Uncertainty Tracker`	0.146139630
`Infectious Disease EMV Tracker`	0.116081938
`Macroeconomic News and Outlook EMV Tracker`	0.085038401
`Macro - Broad Quantity Indicators EMV Tracker`	0.194209264
`Macro - Inflation EMV Indicator`	0.266552837
`Macro - Interest Rates EMV Tracker`	-0.487245155
`Macro - Other Financial Indicators EMV Tracker`	-0.005859121
`Macro - Labor Markets EMV Tracker`	0.193283394
`Macro - Real Estate Markets EMV Tracker`	0.022784693
`Macro - Trade EMV Tracker`	-1.675209154
`Macro - Business Investment and Sentiment EMV Tracker`	0.018843030
`Macro - Consumer Spending and Sentiment EMV Tracker`	0.229134887
`Commodity Markets EMV Tracker`	0.124541757
`Financial Crises EMV Tracker`	0.039697755
`Exchange Rates EMV Tracker`	0.002452335
`Healthcare Matters EMV Tracker`	-0.222996181
`Litigation Matters EMV Tracker`	-0.683592598
`Competition Matters EMV Tracker`	0.361183470
`Labor Disputes EMV Tracker`	2.040024950
`Intellectual Property Matters EMV Tracker`	1.595695423
`Fiscal Policy EMV Tracker`	0.040033614
`Taxes EMV Tracker`	-0.018983435
`Government Spending, Deficits, and Debt EMV Tracker`	-0.400815572
`Entitlement and Welfare Programs EMV Tracker`	0.449153304
`Monetary Policy EMV Tracker`	-0.157550183
`Regulation EMV Tracker`	0.194569410
`Financial Regulation EMV Tracker`	0.689338103
`Competition Policy EMV Tracker`	0.776201330
`Intellectual Property Policy EMV Tracker`	0.882517274
`Labor Regulations EMV Tracker`	-0.517248332
`Immigration EMV Tracker`	-2.296213227
`Energy and Environmental Regulation EMV Tracker`	2.738748976

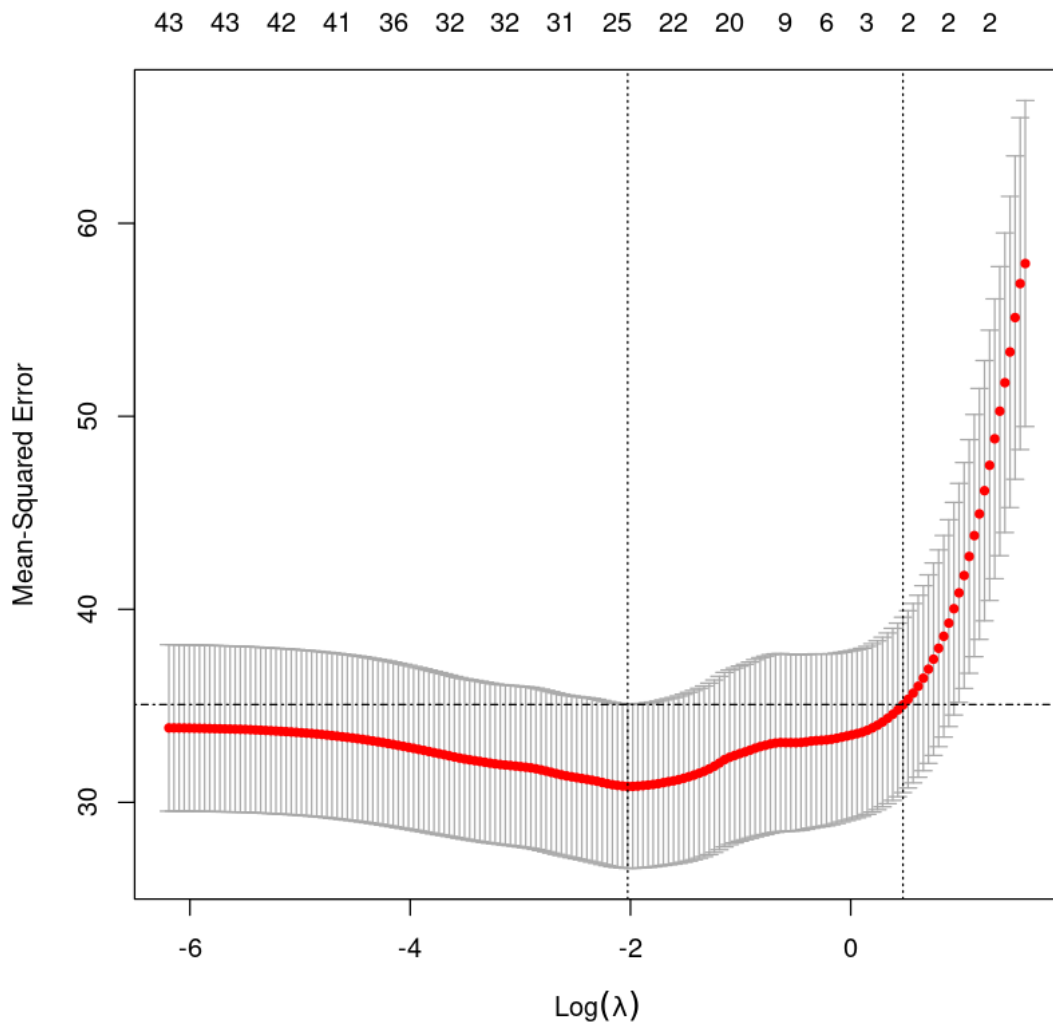
`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	-0.520322102
`Housing and Land Management EMV Tracker`	0.197905224
`Other Regulation EMV Tracker`	-1.492278422
`National Security Policy EMV Tracker`	0.080947034
`Government-Sponsored Enterprises EMV Tracker`	0.462770984
`Trade Policy EMV Tracker`	-0.472899227
`Healthcare Policy EMV Tracker`	-0.477963608
`Food and Drug Policy EMV Tracker`	0.638274589
`Transportation, Infrastructure, and Public Utilities EMV Tracker`	0.491215377
`Elections and Political Governance EMV Tracker`	-0.238530060
`Agricultural Policy EMV Tracker`	-4.903301318
`Petroleum Markets EMV Tracker`	0.021292871

Lasso Regression

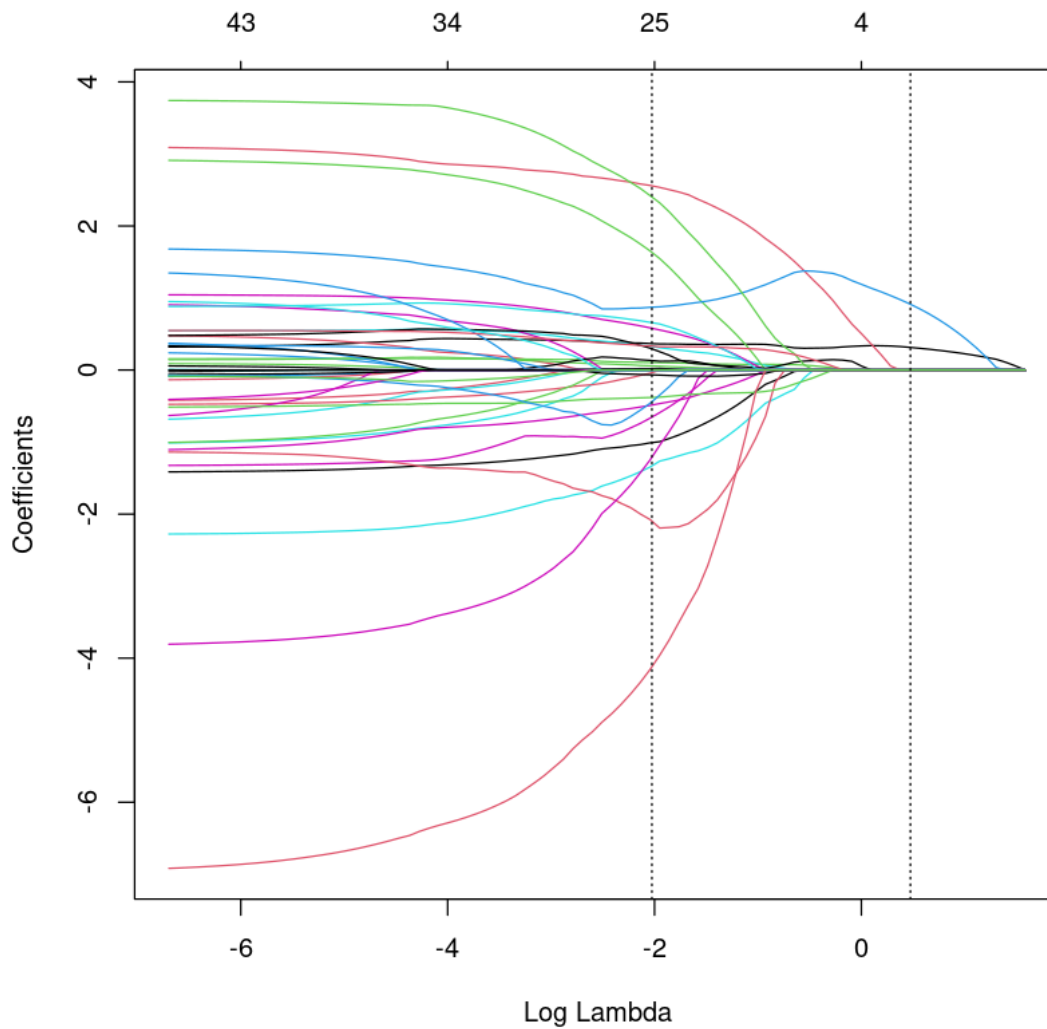
```
[18]: set.seed(1)
lasso.cv = cv.glmnet(x, y, type.measure="mse", alpha=1, family="gaussian",
  ↪ nlambda=200, nfolds = 10)
lasso.cv$lambda.min
lasso.cv$lambda.1se
plot(lasso.cv)
abline( h = lasso.cv$cvup[lasso.cv$index[1]], lty = 4 )
```

0.131970779151975

1.60658947105959



```
[19]: lasso = glmnet(x, y, alpha = 1)
      plot(lasso, xvar = 'lambda')
      abline(v = log(lasso.cv$lambda.min), lty = 3) # careful to use the log here
      abline(v = log(lasso.cv$lambda.1se), lty = 3)
```



Finding MSE, R Squared, RMSE, VIF

```
[20]: lasso = glmnet(x, y, alpha = 1)
predicted_values = predict(lasso, newx = x, s = lasso.cv$lambda.min)
MSE = mean((y - predicted_values)^2)
MSE
SSE = sum((y - predicted_values)^2)
SST = sum((y - mean(y))^2)
R_squared = 1 - (SSE/SST)
R_squared
RMSE = RMSE(predicted_values, y)
RMSE
VIF = 1/(1-R_squared)
```

VIF

23.7447105464929

0.59154932670567

4.87285445570591

2.44827604747121

Check Coefficients

[21]: `coef(lasso, lasso.cv$lambda.min)`

46 x 1 sparse Matrix of class "dgCMatrix"

	s1
(Intercept)	8.33839458
EMV	0.36686824
`Political Uncertainty Tracker`	.
`Infectious Disease EMV Tracker`	0.10419640
`Macroeconomic News and Outlook EMV Tracker`	.
`Macro - Broad Quantity Indicators EMV Tracker`	0.32772695
`Macro - Inflation EMV Indicator`	0.58002343
`Macro - Interest Rates EMV Tracker`	-1.00765211
`Macro - Other Financial Indicators EMV Tracker`	.
`Macro - Labor Markets EMV Tracker`	0.05754084
`Macro - Real Estate Markets EMV Tracker`	.
`Macro - Trade EMV Tracker`	-1.33127939
`Macro - Business Investment and Sentiment EMV Tracker`	.
`Macro - Consumer Spending and Sentiment EMV Tracker`	0.13151521
`Commodity Markets EMV Tracker`	.
`Financial Crises EMV Tracker`	.
`Exchange Rates EMV Tracker`	.
`Healthcare Matters EMV Tracker`	.
`Litigation Matters EMV Tracker`	-0.65349973
`Competition Matters EMV Tracker`	.
`Labor Disputes EMV Tracker`	2.55776071
`Intellectual Property Matters EMV Tracker`	1.63363327
`Fiscal Policy EMV Tracker`	.
`Taxes EMV Tracker`	.
`Government Spending, Deficits, and Debt EMV Tracker`	-0.48693041
`Entitlement and Welfare Programs EMV Tracker`	0.29490858
`Monetary Policy EMV Tracker`	-0.03721664
`Regulation EMV Tracker`	.
`Financial Regulation EMV Tracker`	0.87208918
`Competition Policy EMV Tracker`	0.65764097
`Intellectual Property Policy EMV Tracker`	.
`Labor Regulations EMV Tracker`	.
`Immigration EMV Tracker`	-2.09753874
`Energy and Environmental Regulation EMV Tracker`	2.40283940

```

`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`      .
`Housing and Land Management EMV Tracker`                            .
`Other Regulation EMV Tracker`                                       -1.21108701
`National Security Policy EMV Tracker`                               0.01964979
`Government-Sponsored Enterprises EMV Tracker`                       0.33209141
`Trade Policy EMV Tracker`                                           -0.38070598
`Healthcare Policy EMV Tracker`                                      -0.44049994
`Food and Drug Policy EMV Tracker`                                    .
`Transportation, Infrastructure, and Public Utilities EMV Tracker`    .
`Elections and Political Governance EMV Tracker`                     -0.06683444
`Agricultural Policy EMV Tracker`                                    -4.12088572
`Petroleum Markets EMV Tracker`                                       .

```

Elastic Net Regression

```

[22]: options(warn=-1)
      k <- 10
      set.seed(1)
      train_control <- trainControl(method = "cv", number = k)
      elastic_model <- train(y ~ .,
                             data = cbind(x, y),
                             method = "glmnet",
                             preProcess = c("center", "scale"),
                             tuneLength = 10,
                             trControl = train_control)
      print(elastic_model)

```

glmnet

395 samples

45 predictor

Pre-processing: centered (45), scaled (45)

Resampling: Cross-Validated (10 fold)

Summary of sample sizes: 356, 355, 355, 356, 355, 355, ...

Resampling results across tuning parameters:

alpha	lambda	RMSE	Rsquared	MAE
0.1	0.002254153	5.529511	0.4682945	4.105293
0.1	0.005207385	5.529227	0.4683297	4.105039
0.1	0.012029735	5.522111	0.4691613	4.099010
0.1	0.027790248	5.497642	0.4718137	4.080227
0.1	0.064199078	5.458432	0.4760671	4.054256
0.1	0.148308197	5.405722	0.4816033	4.014840
0.1	0.342611170	5.359671	0.4861185	3.986855
0.1	0.791476239	5.330008	0.4914422	4.002323
0.1	1.828412767	5.421746	0.4834244	4.075087
0.1	4.223870637	5.609669	0.4602630	4.253041

0.2	0.002254153	5.536681	0.4674457	4.109940
0.2	0.005207385	5.532379	0.4679533	4.105893
0.2	0.012029735	5.517854	0.4695316	4.093393
0.2	0.027790248	5.487875	0.4727987	4.070962
0.2	0.064199078	5.440674	0.4779750	4.040221
0.2	0.148308197	5.387823	0.4825608	3.996619
0.2	0.342611170	5.322910	0.4897978	3.975612
0.2	0.791476239	5.353002	0.4894821	4.027518
0.2	1.828412767	5.543033	0.4670315	4.176847
0.2	4.223870637	5.742614	0.4487217	4.400992
0.3	0.002254153	5.536416	0.4676337	4.107583
0.3	0.005207385	5.532614	0.4679927	4.104298
0.3	0.012029735	5.514975	0.4698594	4.089592
0.3	0.027790248	5.479959	0.4736174	4.064967
0.3	0.064199078	5.424822	0.4795232	4.027460
0.3	0.148308197	5.375362	0.4827016	3.985962
0.3	0.342611170	5.308149	0.4916452	3.978920
0.3	0.791476239	5.401017	0.4847777	4.062555
0.3	1.828412767	5.615016	0.4587558	4.258198
0.3	4.223870637	5.860757	0.4461710	4.518907
0.4	0.002254153	5.536526	0.4675969	4.108235
0.4	0.005207385	5.529733	0.4683075	4.101830
0.4	0.012029735	5.509927	0.4703823	4.084935
0.4	0.027790248	5.471177	0.4746060	4.058313
0.4	0.064199078	5.412001	0.4806464	4.014358
0.4	0.148308197	5.357498	0.4840157	3.978557
0.4	0.342611170	5.315538	0.4911023	3.993758
0.4	0.791476239	5.484262	0.4734638	4.122283
0.4	1.828412767	5.667685	0.4544887	4.320374
0.4	4.223870637	5.991102	0.4444682	4.635824
0.5	0.002254153	5.538268	0.4673773	4.108704
0.5	0.005207385	5.529803	0.4682547	4.100438
0.5	0.012029735	5.506331	0.4707537	4.081095
0.5	0.027790248	5.462979	0.4755069	4.052322
0.5	0.064199078	5.402533	0.4812194	4.004675
0.5	0.148308197	5.334966	0.4866257	3.968872
0.5	0.342611170	5.320632	0.4917300	4.000644
0.5	0.791476239	5.533611	0.4670476	4.167735
0.5	1.828412767	5.698640	0.4544953	4.361739
0.5	4.223870637	6.132781	0.4491760	4.747241
0.6	0.002254153	5.537604	0.4675092	4.108132
0.6	0.005207385	5.527521	0.4685007	4.098654
0.6	0.012029735	5.502252	0.4711758	4.077975
0.6	0.027790248	5.453320	0.4764483	4.047450
0.6	0.064199078	5.395938	0.4812568	3.995727
0.6	0.148308197	5.317382	0.4887401	3.963194
0.6	0.342611170	5.336651	0.4907621	4.010206
0.6	0.791476239	5.555466	0.4657169	4.196312

0.6	1.828412767	5.733869	0.4536194	4.400348
0.6	4.223870637	6.301184	0.4526365	4.876820
0.7	0.002254153	5.533882	0.4679426	4.104827
0.7	0.005207385	5.525119	0.4687988	4.097017
0.7	0.012029735	5.496595	0.4718635	4.073870
0.7	0.027790248	5.443786	0.4774154	4.041179
0.7	0.064199078	5.389487	0.4811410	3.987365
0.7	0.148308197	5.308649	0.4899024	3.961916
0.7	0.342611170	5.362128	0.4884348	4.026285
0.7	0.791476239	5.572329	0.4644704	4.217133
0.7	1.828412767	5.771306	0.4545160	4.441671
0.7	4.223870637	6.486336	0.4585595	5.014636
0.8	0.002254153	5.531654	0.4680378	4.102422
0.8	0.005207385	5.523872	0.4688028	4.095272
0.8	0.012029735	5.493196	0.4721435	4.070819
0.8	0.027790248	5.436051	0.4781367	4.034486
0.8	0.064199078	5.385971	0.4805885	3.984669
0.8	0.148308197	5.305698	0.4903483	3.965207
0.8	0.342611170	5.397627	0.4844153	4.049245
0.8	0.791476239	5.589100	0.4641517	4.239614
0.8	1.828412767	5.798596	0.4601194	4.475048
0.8	4.223870637	6.689323	0.4578697	5.158876
0.9	0.002254153	5.539105	0.4671769	4.109314
0.9	0.005207385	5.524635	0.4686289	4.095638
0.9	0.012029735	5.489784	0.4724700	4.068738
0.9	0.027790248	5.427974	0.4789956	4.027255
0.9	0.064199078	5.380300	0.4806476	3.981880
0.9	0.148308197	5.305194	0.4908186	3.970404
0.9	0.342611170	5.437453	0.4795360	4.075467
0.9	0.791476239	5.602646	0.4637358	4.256229
0.9	1.828412767	5.827550	0.4666166	4.504222
0.9	4.223870637	6.895660	0.4446532	5.311795
1.0	0.002254153	5.538558	0.4672526	4.108358
1.0	0.005207385	5.522228	0.4689491	4.093683
1.0	0.012029735	5.484366	0.4731303	4.065363
1.0	0.027790248	5.420199	0.4797842	4.019936
1.0	0.064199078	5.370312	0.4814733	3.978184
1.0	0.148308197	5.302217	0.4918332	3.973038
1.0	0.342611170	5.469495	0.4754546	4.098821
1.0	0.791476239	5.609394	0.4642015	4.266156
1.0	1.828412767	5.874034	0.4684257	4.540014
1.0	4.223870637	7.104318	0.4385096	5.475839

RMSE was used to select the optimal model using the smallest value.

The final values used for the model were alpha = 1 and lambda = 0.1483082.

```
[23]: predictions_enet <- glmnet(x, y, alpha = 0,
  ↪elastic_model$finalModel$tuneValue$alpha,
  lambda = elastic_model$finalModel$tuneValue$lambda) %>%
  predict(x) %>% as.vector()
  RMSE_enet = RMSE(predictions_enet, y)
  RMSE_enet
```

4.89902410550464

```
[24]: options(warn=-1)
  k <- 10
  set.seed(1)
  train_control <- trainControl(method = "cv", number = k)
  elastic_model <- train(y ~ .,
  data = cbind(x, y),
  method = "glmnet",
  preProcess = c("center", "scale"),
  tuneLength = 100,
  trControl = train_control)
```

```
[25]: print(elastic_model$finalModel$tuneValue)
```

```
      alpha      lambda
8965 0.9909091 0.1611272
```

```
[41]: predictions_enet <- glmnet(x, y, alpha = 0,
  ↪elastic_model$finalModel$tuneValue$alpha,
  lambda = elastic_model$finalModel$tuneValue$lambda) %>%
  predict(x) %>% as.vector()

  MSE = mean((y - predicted_values)^2)
  MSE

  RMSE_enet = RMSE(predictions_enet, y)
  RMSE_enet

  SSE = sum((y-predictions_enet)^2)
  SST = sum((y-mean(y))^2)
  r_sq = 1-(SSE/SST)
  r_sq
  vif = 1/(1-r_sq)
  vif
```

22.0113727453303

5.28824015974154

0.518944619637773

2.07876273880778

Phase 2

Segment 1

```
[33]: all_data <- read_excel("EMV_VIX_Data.xlsx")
      selected_data <- all_data[1:102,]

      y_set <- selected_data$VIX
      data <- all_data[,!(names(all_data)%in%c("VIX", "Date"))]
      selected_data <- data[1:102, ]
      x_set <- selected_data$EMV

      x_matrix <- scale(model.matrix(y_set~., selected_data)[,-1])

      options(warn=-1)
      k <- 102
      set.seed(1)
      train_control <- trainControl(method = "loocv", number = k)
      elastic_model <- train(y_set ~ .,
        data = cbind(x_matrix, y_set),
        method = "glmnet",
        preprocess = c("center", "scale"),
        tuneLength = 10,
        trControl = train_control)
      enet = glmnet(x_matrix, y_set, alpha = elastic_model$finalModel$tuneValue$alpha)
      predicted_values <- predict(enet, newx =
        ↪x_matrix,s=elastic_model$finalModel$tuneValue$lambda)
      coef(enet, elastic_model$finalModel$tuneValue$lambda)
      print(elastic_model$finalModel$tuneValue)
```

46 x 1 sparse Matrix of class "dgCMatrix"

	s1
(Intercept)	17.07990191
EMV	.
`Political Uncertainty Tracker`	.
`Infectious Disease EMV Tracker`	.
`Macroeconomic News and Outlook EMV Tracker`	.
`Macro - Broad Quantity Indicators EMV Tracker`	.
`Macro - Inflation EMV Indicator`	.
`Macro - Interest Rates EMV Tracker`	-0.95748740
`Macro - Other Financial Indicators EMV Tracker`	0.16067622
`Macro - Labor Markets EMV Tracker`	.
`Macro - Real Estate Markets EMV Tracker`	0.52205143
`Macro - Trade EMV Tracker`	-0.29218732
`Macro - Business Investment and Sentiment EMV Tracker`	.
`Macro - Consumer Spending and Sentiment EMV Tracker`	.
`Commodity Markets EMV Tracker`	1.13051068

`Financial Crises EMV Tracker`	.
`Exchange Rates EMV Tracker`	.
`Healthcare Matters EMV Tracker`	-0.37446126
`Litigation Matters EMV Tracker`	-0.45997446
`Competition Matters EMV Tracker`	.
`Labor Disputes EMV Tracker`	0.64191309
`Intellectual Property Matters EMV Tracker`	.
`Fiscal Policy EMV Tracker`	.
`Taxes EMV Tracker`	.
`Government Spending, Deficits, and Debt EMV Tracker`	-0.23048151
`Entitlement and Welfare Programs EMV Tracker`	.
`Monetary Policy EMV Tracker`	.
`Regulation EMV Tracker`	.
`Financial Regulation EMV Tracker`	0.74082823
`Competition Policy EMV Tracker`	.
`Intellectual Property Policy EMV Tracker`	.
`Labor Regulations EMV Tracker`	0.90510298
`Immigration EMV Tracker`	.
`Energy and Environmental Regulation EMV Tracker`	0.10046558
`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	.
`Housing and Land Management EMV Tracker`	.
`Other Regulation EMV Tracker`	.
`National Security Policy EMV Tracker`	.
`Government-Sponsored Enterprises EMV Tracker`	-0.29025937
`Trade Policy EMV Tracker`	-0.87404599
`Healthcare Policy EMV Tracker`	0.20626213
`Food and Drug Policy EMV Tracker`	.
`Transportation, Infrastructure, and Public Utilities EMV Tracker`	.
`Elections and Political Governance EMV Tracker`	-0.05130814
`Agricultural Policy EMV Tracker`	.
`Petroleum Markets EMV Tracker`	0.88149912

alpha lambda

48 0.5 0.4040454

Segment 2

```
[34]: all_data <- read_excel("EMV_VIX_Data.xlsx")
      selected_data <- all_data[104:159,]

      y_set <- selected_data$VIX
      data <- all_data[,!(names(all_data)%in%c("VIX", "Date"))]
      selected_data <- data[104:159, ]
      x_set <- selected_data$EMV

      x_matrix <- scale(model.matrix(y_set~., selected_data)[,-1])

      options(warn=-1)
```

```

k <- 55
set.seed(1)
train_control <- trainControl(method = "loocv", number = k)
elastic_model <- train(y_set ~ .,
data = cbind(x_matrix, y_set),
method = "glmnet",
preProcess = c("center", "scale"),
tuneLength = 10,
trControl = train_control)
enet = glmnet(x_matrix, y_set, alpha = elastic_model$finalModel$tuneValue$alpha)
predicted_values <- predict(enet, newx =
  ↪x_matrix,s=elastic_model$finalModel$tuneValue$lambda)
coef(enet, elastic_model$finalModel$tuneValue$lambda)
print(elastic_model$finalModel$tuneValue)

```

46 x 1 sparse Matrix of class "dgCMatrix"

	s1
(Intercept)	25.53571420
EMV	.
`Political Uncertainty Tracker`	.
`Infectious Disease EMV Tracker`	.
`Macroeconomic News and Outlook EMV Tracker`	.
`Macro - Broad Quantity Indicators EMV Tracker`	.
`Macro - Inflation EMV Indicator`	.
`Macro - Interest Rates EMV Tracker`	.
`Macro - Other Financial Indicators EMV Tracker`	.
`Macro - Labor Markets EMV Tracker`	.
`Macro - Real Estate Markets EMV Tracker`	.
`Macro - Trade EMV Tracker`	0.50874215
`Macro - Business Investment and Sentiment EMV Tracker`	0.99933893
`Macro - Consumer Spending and Sentiment EMV Tracker`	.
`Commodity Markets EMV Tracker`	.
`Financial Crises EMV Tracker`	.
`Exchange Rates EMV Tracker`	.
`Healthcare Matters EMV Tracker`	0.02967842
`Litigation Matters EMV Tracker`	.
`Competition Matters EMV Tracker`	.
`Labor Disputes EMV Tracker`	0.90402629
`Intellectual Property Matters EMV Tracker`	.
`Fiscal Policy EMV Tracker`	.
`Taxes EMV Tracker`	.
`Government Spending, Deficits, and Debt EMV Tracker`	.
`Entitlement and Welfare Programs EMV Tracker`	.
`Monetary Policy EMV Tracker`	.
`Regulation EMV Tracker`	.
`Financial Regulation EMV Tracker`	0.23025611
`Competition Policy EMV Tracker`	.

```

`Intellectual Property Policy EMV Tracker`      .
`Labor Regulations EMV Tracker`                  .
`Immigration EMV Tracker`                        .
`Energy and Environmental Regulation EMV Tracker` .
`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker` .
`Housing and Land Management EMV Tracker`        .
`Other Regulation EMV Tracker`                   0.39371631
`National Security Policy EMV Tracker`           .
`Government-Sponsored Enterprises EMV Tracker`   0.02150630
`Trade Policy EMV Tracker`                       0.71297125
`Healthcare Policy EMV Tracker`                  .
`Food and Drug Policy EMV Tracker`               .
`Transportation, Infrastructure, and Public Utilities EMV Tracker` .
`Elections and Political Governance EMV Tracker` .
`Agricultural Policy EMV Tracker`                .
`Petroleum Markets EMV Tracker`                  .

```

```

      alpha      lambda
49    0.5 0.9957554

```

Segment 3

```

[35]: all_data <- read_excel("EMV_VIX_Data.xlsx")
      selected_data <- all_data[161:216,]

      y_set <- selected_data$VIX
      data <- all_data[,!(names(all_data)%in%c("VIX", "Date"))]
      selected_data <- data[161:216, ]
      x_set <- selected_data$EMV

      x_matrix <- scale(model.matrix(y_set~., selected_data)[,-1])

      options(warn=-1)
      k <- 55
      set.seed(1)
      train_control <- trainControl(method = "loocv", number = k)
      elastic_model <- train(y_set ~ .,
                             data = cbind(x_matrix, y_set),
                             method = "glmnet",
                             preprocess = c("center", "scale"),
                             tuneLength = 10,
                             trControl = train_control)
      enet = glmnet(x_matrix, y_set, alpha = elastic_model$finalModel$tuneValue$alpha)
      predicted_values <- predict(enet, newx =
        ↪ x_matrix,s=elastic_model$finalModel$tuneValue$lambda)
      coef(enet, elastic_model$finalModel$tuneValue$lambda)
      print(elastic_model$finalModel$tuneValue)

```

46 x 1 sparse Matrix of class "dgCMatrix"

	s1
(Intercept)	15.311964286
EMV	0.039221228
`Political Uncertainty Tracker`	0.046505225
`Infectious Disease EMV Tracker`	0.119161854
`Macroeconomic News and Outlook EMV Tracker`	0.055469282
`Macro - Broad Quantity Indicators EMV Tracker`	0.100430569
`Macro - Inflation EMV Indicator`	-0.180788567
`Macro - Interest Rates EMV Tracker`	.
`Macro - Other Financial Indicators EMV Tracker`	0.395524300
`Macro - Labor Markets EMV Tracker`	.
`Macro - Real Estate Markets EMV Tracker`	0.289676099
`Macro - Trade EMV Tracker`	-0.291921404
`Macro - Business Investment and Sentiment EMV Tracker`	.
`Macro - Consumer Spending and Sentiment EMV Tracker`	.
`Commodity Markets EMV Tracker`	.
`Financial Crises EMV Tracker`	.
`Exchange Rates EMV Tracker`	.
`Healthcare Matters EMV Tracker`	.
`Litigation Matters EMV Tracker`	.
`Competition Matters EMV Tracker`	.
`Labor Disputes EMV Tracker`	0.192284428
`Intellectual Property Matters EMV Tracker`	-0.172895575
`Fiscal Policy EMV Tracker`	0.673867565
`Taxes EMV Tracker`	0.597598983
`Government Spending, Deficits, and Debt EMV Tracker`	.
`Entitlement and Welfare Programs EMV Tracker`	-0.343877228
`Monetary Policy EMV Tracker`	.
`Regulation EMV Tracker`	.
`Financial Regulation EMV Tracker`	-0.219155093
`Competition Policy EMV Tracker`	.
`Intellectual Property Policy EMV Tracker`	-0.063458848
`Labor Regulations EMV Tracker`	.
`Immigration EMV Tracker`	.
`Energy and Environmental Regulation EMV Tracker`	0.550065936
`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	.
`Housing and Land Management EMV Tracker`	.
`Other Regulation EMV Tracker`	0.358387121
`National Security Policy EMV Tracker`	0.086345765
`Government-Sponsored Enterprises EMV Tracker`	0.149879629
`Trade Policy EMV Tracker`	.
`Healthcare Policy EMV Tracker`	-0.226839815
`Food and Drug Policy EMV Tracker`	0.178997241
`Transportation, Infrastructure, and Public Utilities EMV Tracker`	0.001355857
`Elections and Political Governance EMV Tracker`	-0.148939699
`Agricultural Policy EMV Tracker`	.
`Petroleum Markets EMV Tracker`	-0.123830392


```

alpha    lambda
10      0.1 1.697438

```

Segment 4

```

[36]: all_data <- read_excel("EMV_VIX_Data.xlsx")
selected_data <- all_data[218:237,]

y_set <- selected_data$VIX
data <- all_data[,!(names(all_data)%in%c("VIX", "Date"))]
selected_data <- data[218:237, ]
x_set <- selected_data$EMV

x_matrix <- scale(model.matrix(y_set~., selected_data)[,-1])

options(warn=-1)
k <- 19
set.seed(1)
train_control <- trainControl(method = "loocv", number = k)
elastic_model <- train(y_set ~ .,
data = cbind(x_matrix, y_set),
method = "glmnet",
preProcess = c("center", "scale"),
tuneLength = 10,
trControl = train_control)
enet = glmnet(x_matrix, y_set, alpha = elastic_model$finalModel$tuneValue$alpha)
predicted_values <- predict(enet, newx =
  ↪x_matrix,s=elastic_model$finalModel$tuneValue$lambda)
coef(enet, elastic_model$finalModel$tuneValue$lambda)
print(elastic_model$finalModel$tuneValue)

```

46 x 1 sparse Matrix of class "dgCMatrix"

	s1
(Intercept)	33.083000000
EMV	0.304650826
`Political Uncertainty Tracker`	0.204953542
`Infectious Disease EMV Tracker`	.
`Macroeconomic News and Outlook EMV Tracker`	0.190354606
`Macro - Broad Quantity Indicators EMV Tracker`	0.344339897
`Macro - Inflation EMV Indicator`	.
`Macro - Interest Rates EMV Tracker`	.
`Macro - Other Financial Indicators EMV Tracker`	.
`Macro - Labor Markets EMV Tracker`	0.950868943
`Macro - Real Estate Markets EMV Tracker`	.
`Macro - Trade EMV Tracker`	.
`Macro - Business Investment and Sentiment EMV Tracker`	.
`Macro - Consumer Spending and Sentiment EMV Tracker`	.
`Commodity Markets EMV Tracker`	.

`Financial Crises EMV Tracker`	0.717122105
`Exchange Rates EMV Tracker`	.
`Healthcare Matters EMV Tracker`	0.540386691
`Litigation Matters EMV Tracker`	.
`Competition Matters EMV Tracker`	.
`Labor Disputes EMV Tracker`	.
`Intellectual Property Matters EMV Tracker`	0.351182806
`Fiscal Policy EMV Tracker`	0.594512256
`Taxes EMV Tracker`	0.532015752
`Government Spending, Deficits, and Debt EMV Tracker`	0.849545095
`Entitlement and Welfare Programs EMV Tracker`	0.384427990
`Monetary Policy EMV Tracker`	.
`Regulation EMV Tracker`	0.447005158
`Financial Regulation EMV Tracker`	0.560023094
`Competition Policy EMV Tracker`	.
`Intellectual Property Policy EMV Tracker`	0.393961006
`Labor Regulations EMV Tracker`	0.370724123
`Immigration EMV Tracker`	0.411338452
`Energy and Environmental Regulation EMV Tracker`	.
`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	.
`Housing and Land Management EMV Tracker`	.
`Other Regulation EMV Tracker`	-0.970109763
`National Security Policy EMV Tracker`	0.279376978
`Government-Sponsored Enterprises EMV Tracker`	0.468289890
`Trade Policy EMV Tracker`	0.338393100
`Healthcare Policy EMV Tracker`	0.553653792
`Food and Drug Policy EMV Tracker`	.
`Transportation, Infrastructure, and Public Utilities EMV Tracker`	0.004744626
`Elections and Political Governance EMV Tracker`	.
`Agricultural Policy EMV Tracker`	.
`Petroleum Markets EMV Tracker`	.

```

alpha    lambda
10      0.1 12.59202

```

Segment 5

```

[37]: all_data <- read_excel("EMV_VIX_Data.xlsx")
      selected_data <- all_data[239:360,]

      y_set <- selected_data$VIX
      data <- all_data[,!(names(all_data)%in%c("VIX", "Date"))]
      selected_data <- data[239:360, ]
      x_set <- selected_data$EMV

      x_matrix <- scale(model.matrix(y_set~., selected_data)[,-1])

      options(warn=-1)

```

```

k <- 21
set.seed(1)
train_control <- trainControl(method = "loocv", number = k)
elastic_model <- train(y_set ~ .,
data = cbind(x_matrix, y_set),
method = "glmnet",
preProcess = c("center", "scale"),
tuneLength = 10,
trControl = train_control)
enet = glmnet(x_matrix, y_set, alpha = elastic_model$finalModel$tuneValue$alpha)
predicted_values <- predict(enet, newx =
  ↪x_matrix,s=elastic_model$finalModel$tuneValue$lambda)
coef(enet, elastic_model$finalModel$tuneValue$lambda)
print(elastic_model$finalModel$tuneValue)

```

46 x 1 sparse Matrix of class "dgCMatrix"

	s1
(Intercept)	17.13893440
EMV	.
`Political Uncertainty Tracker`	.
`Infectious Disease EMV Tracker`	-0.42291170
`Macroeconomic News and Outlook EMV Tracker`	.
`Macro - Broad Quantity Indicators EMV Tracker`	.
`Macro - Inflation EMV Indicator`	.
`Macro - Interest Rates EMV Tracker`	.
`Macro - Other Financial Indicators EMV Tracker`	0.23403989
`Macro - Labor Markets EMV Tracker`	.
`Macro - Real Estate Markets EMV Tracker`	2.50387975
`Macro - Trade EMV Tracker`	0.29491006
`Macro - Business Investment and Sentiment EMV Tracker`	.
`Macro - Consumer Spending and Sentiment EMV Tracker`	.
`Commodity Markets EMV Tracker`	.
`Financial Crises EMV Tracker`	.
`Exchange Rates EMV Tracker`	.
`Healthcare Matters EMV Tracker`	-0.69059462
`Litigation Matters EMV Tracker`	-0.14324169
`Competition Matters EMV Tracker`	0.24761477
`Labor Disputes EMV Tracker`	0.14578319
`Intellectual Property Matters EMV Tracker`	0.42146304
`Fiscal Policy EMV Tracker`	.
`Taxes EMV Tracker`	.
`Government Spending, Deficits, and Debt EMV Tracker`	.
`Entitlement and Welfare Programs EMV Tracker`	0.14714266
`Monetary Policy EMV Tracker`	.
`Regulation EMV Tracker`	.
`Financial Regulation EMV Tracker`	1.29428527
`Competition Policy EMV Tracker`	.

`Intellectual Property Policy EMV Tracker`	0.22995490
`Labor Regulations EMV Tracker`	-0.07204736
`Immigration EMV Tracker`	-0.54304675
`Energy and Environmental Regulation EMV Tracker`	.
`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	.
`Housing and Land Management EMV Tracker`	-0.10076196
`Other Regulation EMV Tracker`	.
`National Security Policy EMV Tracker`	.
`Government-Sponsored Enterprises EMV Tracker`	.
`Trade Policy EMV Tracker`	.
`Healthcare Policy EMV Tracker`	.
`Food and Drug Policy EMV Tracker`	.
`Transportation, Infrastructure, and Public Utilities EMV Tracker`	.
`Elections and Political Governance EMV Tracker`	-0.27156557
`Agricultural Policy EMV Tracker`	-0.26248949
`Petroleum Markets EMV Tracker`	0.37344208

```

alpha    lambda
57 0.6 0.2840449

```

Segment 6

```

[38]: all_data <- read_excel("EMV_VIX_Data.xlsx")
      selected_data <- all_data[350:395,]

      y_set <- selected_data$VIX
      data <- all_data[,!(names(all_data)%in%c("VIX", "Date"))]
      selected_data <- data[350:395, ]
      x_set <- selected_data$EMV

      x_matrix <- scale(model.matrix(y_set~., selected_data)[,-1])

      options(warn=-1)
      k <- 45
      set.seed(1)
      train_control <- trainControl(method = "loocv", number = k)
      elastic_model <- train(y_set ~ .,
                             data = cbind(x_matrix, y_set),
                             method = "glmnet",
                             preprocess = c("center", "scale"),
                             tuneLength = 10,
                             trControl = train_control)
      enet = glmnet(x_matrix, y_set, alpha = elastic_model$finalModel$tuneValue$alpha)
      predicted_values <- predict(enet, newx =
        ↪ x_matrix,s=elastic_model$finalModel$tuneValue$lambda)
      coef(enet, elastic_model$finalModel$tuneValue$lambda)
      print(elastic_model$finalModel$tuneValue)

```

46 x 1 sparse Matrix of class "dgCMatrix"

	s1
(Intercept)	23.384782761
EMV	.
`Political Uncertainty Tracker`	.
`Infectious Disease EMV Tracker`	1.219542947
`Macroeconomic News and Outlook EMV Tracker`	.
`Macro - Broad Quantity Indicators EMV Tracker`	0.327933056
`Macro - Inflation EMV Indicator`	0.585607942
`Macro - Interest Rates EMV Tracker`	.
`Macro - Other Financial Indicators EMV Tracker`	-0.898253234
`Macro - Labor Markets EMV Tracker`	.
`Macro - Real Estate Markets EMV Tracker`	-0.075879125
`Macro - Trade EMV Tracker`	-0.424815430
`Macro - Business Investment and Sentiment EMV Tracker`	-0.712603686
`Macro - Consumer Spending and Sentiment EMV Tracker`	0.234450322
`Commodity Markets EMV Tracker`	0.821882583
`Financial Crises EMV Tracker`	.
`Exchange Rates EMV Tracker`	0.734751258
`Healthcare Matters EMV Tracker`	.
`Litigation Matters EMV Tracker`	0.002058396
`Competition Matters EMV Tracker`	.
`Labor Disputes EMV Tracker`	-0.318700798
`Intellectual Property Matters EMV Tracker`	0.560189520
`Fiscal Policy EMV Tracker`	-0.488006112
`Taxes EMV Tracker`	-1.171158070
`Government Spending, Deficits, and Debt EMV Tracker`	-0.148743921
`Entitlement and Welfare Programs EMV Tracker`	.
`Monetary Policy EMV Tracker`	.
`Regulation EMV Tracker`	-0.408828824
`Financial Regulation EMV Tracker`	0.012116995
`Competition Policy EMV Tracker`	0.841038078
`Intellectual Property Policy EMV Tracker`	-0.646522789
`Labor Regulations EMV Tracker`	-1.296741150
`Immigration EMV Tracker`	0.611096236
`Energy and Environmental Regulation EMV Tracker`	1.590652837
`Lawsuit and Tort Reform, Supreme Court Decisions EMV Tracker`	-0.459477404
`Housing and Land Management EMV Tracker`	.
`Other Regulation EMV Tracker`	0.229677458
`National Security Policy EMV Tracker`	0.737461102
`Government-Sponsored Enterprises EMV Tracker`	.
`Trade Policy EMV Tracker`	-1.232626844
`Healthcare Policy EMV Tracker`	.
`Food and Drug Policy EMV Tracker`	1.529258567
`Transportation, Infrastructure, and Public Utilities EMV Tracker`	.
`Elections and Political Governance EMV Tracker`	0.445896613
`Agricultural Policy EMV Tracker`	.
`Petroleum Markets EMV Tracker`	.

```
alpha  lambda
10    0.1 2.69543
```

```
[ ]:
```